

Data Set #6

Type of response:	Non-Source Dependent Response
Grade level:	10
Subject:	Biology
Training set size:	1797
Final evaluation set size:	599
Average length of responses:	50 words
Scoring:	Score1, Score2
Final score:	Final score is score 1. Score 2 is for inter-rater reliability purposes.
Rubric range:	0-3

Prompt—Cell Membrane Item

List and describe three processes used by cells to control the movement of substances across the cell membrane.

Rubric for Cell Membrane

Key Elements:

- Selective permeability is used by the cell membrane to allow certain substances to move across.
- Passive transport occurs when substances move from an area of higher concentration to an area of lower concentration.
- Osmosis is the diffusion of water across the cell membrane.
- Facilitated diffusion occurs when the membrane controls the pathway for a particle to enter or leave a cell.
- Active transport occurs when a cell uses energy to move a substance across the cell membrane, and/or a substance moves from an area of low to high concentration, or against the concentration gradient.
- Pumps are used to move charged particles like sodium and potassium ions through membranes using energy and carrier proteins.
- Membrane-assisted transport occurs when the membrane of the vesicle fuses with the cell membrane forcing large molecules out of the cell as in exocytosis.
- Membrane-assisted transport occurs when molecules are engulfed by the cell membrane as in endocytosis.
- Membrane-assisted transport occurs when vesicles are formed around large molecules as in phagocytosis.
- Membrane-assisted transport occurs when vesicles are formed around liquid droplets as in pinocytosis.
- Protein channels or channel proteins allow for the movement of specific molecules or substances into or out of the cell.

Rubric:

3 points

Three key elements

2 points

Two key elements

1 point
One key element

0 points
Other